Original leg prototype (milestone M3) + The single-leg test rig print queue

Gyropup retained prototype workstreams · Revision: cd276824fcbe0bea6c3c39099120afd465b751a3

Matching parts, all pages: 54 of 54

Original leg, rig, turret and earlier body planning lists. Current front/rear-pack robot manufacturing plan remains incomplete. Checklist marks track parts, not purchased quantities. This list does not qualify fit, strength or a complete robot.

Done	Part ID	Order	Part	Material	Quantity	Grams	Hours	Print setup	Printed
☐	print/rig/pod	1	pod	petg	1	493	19.7	Placement and instructions 230 × 200 × 140 mm · straight PETG, 3 walls, 25 % infill; 4 x M8 x 16 into the post's back slots; 3 x M4 inserts (contactor, relay) 2 mm walls, 40 % gyroid Plan day 1	☑
☐	print/rig/dyno_lug_plate	2	dyno_lug_plate	petg	2	417	16.7	Placement and instructions 150 × 76 × 156 mm · straight PETG, 6 walls, 60 % gyroid; on a 19 mm plywood board, 4 x M6 through bolts each (B1, B2) 2 mm walls, 40 % gyroid Plan day 1	☐
☐	print/leg/stop_sleeve	3	stop_sleeve	tpu95	4	62	6.4	Placement and instructions 32 × 32 × 20 mm · straight printed TPU 95A cap-down, 100 % solid; a stretch fit on the stop pins 100 % Plan day 2	☐
☐	print/leg/upper_link	4	upper_link	pa6-cf	2	848	55.6	Placement and instructions 214 × 94 × 50 mm · straight printed PA6-CF flat, 30 mm, 100 % solid; 8 x M5 into the adapter; the knee shoulder screw clamped by an M10 nyloc in a pocket; a radial stop face each side of the hub 100 % Plan day 3-5	☐
☐	print/leg/front_link	5	front_link	pa6-cf	1	463	30.3	Placement and instructions 320 × 90 × 80 mm · turned printed PA6-CF flat on its inner face, 20 mm body; two 6301-2RS at the knee, two 61812 at the apex (pressed against a central web) 100 % Plan day 5-6	☐
☐	print/leg/wheel_link	6	wheel_link	pa6-cf	1	647	42.4	See note below	☐
Print setup: Placement and instructions 318 × 84 × 86 mm · turned printed PA6-CF flat on its outer face, 26 mm body; two 6301-2RS at the knee; the BotWheel in its outer face (4 x M6 from inside the hub), the hub for the apex 61812 (DIN 471-60 circlip) 100 % Plan day 6-8									

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☐	print/leg/hip_frame	7	hip_frame	pa6-cf	1	1106	72.5	See note below	☐
Print setup: Placement and instructions 274 × 158 × 81 mm · turned printed PA6-CF web-down (turned 45 deg on the bed), 100 % solid; 20 x M4 into the motors, 2 x 61812 pressed in from the motor side onto a 2 mm lip (no retainers), 6 x M5 inserts in the flange ends, 4 x Ø16 stop pins in blind holes (23 deep, epoxy-bonded, D74) 100 % Plan day 8-10									
☐	print/leg/apex_spacer	8	apex_spacer	pa6-cf	1	14	0.9	Placement and instructions 66 × 66 × 22 mm · straight printed PA6-CF ring between the apex 61812s' inner races 100 % Plan day 10	☐
☐	print/rig/bumper_tpu	9	bumper_tpu	tpu95	4	303	31	See note below	☐
Print setup: Placement and instructions 53 × 53 × 40 mm · straight TPU 95A, 30 % gyroid, 2 walls (sized for 33 J: a hard fault at a jump's apex; docs/leg-bench-tests.md R0 confirms the infill by dropping the carriage on them); two under the plate, two in the top stop 2 mm walls, 40 % gyroid									
☐	print/rig/coupon_c4_pr2	10	coupon_c4_pr2	pa6-cf	2	452	29.6	Placement and instructions 118 × 88 × 50 mm · straight PA6-CF, 100 %, dried; outer face down, stub up, as the link (C4: the drawn stub) 100 %	☐
☐	print/rig/coupon_c4_today	11	coupon_c4_today	pa6-cf	2	402	26.3	Placement and instructions 110 × 88 × 50 mm · straight PA6-CF, 100 %, dried; outer face down, stub up, as the link (C4: dev's stub) 100 %	☐
☐	print/rig/press_cage	12	press_cage	petg	1	335	13.4	Placement and instructions 242 × 94 × 50 mm · straight PETG, 6 walls, 40 % gyroid; flat (B3, B4) 2 mm walls, 40 % gyroid	☐
☐	print/rig/dyno_hub	13	dyno_hub	petg	2	292	11.7	Placement and instructions 90 × 90 × 44 mm · straight PETG, 100 %; 9 x M5 x 25 into the output (the leg's D46 screw), 3 pin sockets 100 %	☐
☐	print/rig/coupon_frame_ring_nom	14	coupon_frame_ring_nom	pa6-cf	1	177	11.6	Placement and instructions 124 × 124 × 24 mm · straight PA6-CF, 100 %, dried; the seat at 0 mm on the leg's 100 %	☐
☐	print/rig/drop_bracket	15	drop_bracket	petg	1	199	7.9	Placement and instructions 105 × 100 × 40 mm · straight PETG, 100 %; fitted for the drop test only (2 x M8 x 16 into the post's side slots) 100 %	☐
☐	print/rig/puller_m5	16	puller_m5	pa6-cf	3	115	7.5	Placement and instructions 73 × 30 × 60 mm · straight PA6-CF, 6 walls, 60 % gyroid, insert face up (B4: two to 3.5 kN, one for the head test) 2 mm walls, 40 % gyroid	☐
☐	print/rig/top_stop_bracket	17	top_stop_bracket	petg	1	186	7.4	Placement and instructions 220 × 68 × 66 mm · straight PETG, 6 walls, 50 % gyroid; 4 x M8 x 16 into T-nuts on the post's front 2 mm walls, 40 % gyroid	☐

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☐	print/rig/dyno_spider	18	dyno_spider	tpu95	2	68	7	Placement and instructions 87 × 80 × 20 mm · straight TPU 95A, 100 %; one spare 100 %	☐
☐	print/rig/bumper_cap	19	bumper_cap	petg	4	116	4.7	Placement and instructions 60 × 60 × 13 mm · straight PETG, 100 % (it spreads the plate edge's blow); printed upside down 100 %	☐
☐	print/rig/coupon_c3_sleeved	20	coupon_c3_sleeved	pa6-cf	2	67	4.4	Placement and instructions 60 × 40 × 13 mm · straight PA6-CF, 100 %, dried, flat (C3: an 8 x 5.3 steel sleeve flush, M5 8.8 through the B4 load cell) 100 %	☐
☐	print/rig/coupon_c3_plain	21	coupon_c3_plain	pa6-cf	2	67	4.4	Placement and instructions 60 × 40 × 13 mm · straight PA6-CF, 100 %, dried, flat (C3: M5 8.8 on the plastic, a Ø15 washer) 100 %	☐
☐	print/rig/bumper_base	22	bumper_base	petg	2	105	4.2	Placement and instructions 60 × 60 × 29 mm · straight PETG, 4 walls, 40 % gyroid; 2 x M6 x 45 countersunk through the deck into T-nuts 2 mm walls, 40 % gyroid	☐
☐	print/rig/sleeve_block	23	sleeve_block	pa6-cf	1	53	3.5	Placement and instructions 62 × 40 × 40 mm · straight PA6-CF, 100 %; a real stop post (Ø16 pin) in its blind hole and TPU sleeve (B3) 100 %	☐
☐	print/rig/pod_lid	24	pod_lid	petg	1	65	2.6	Placement and instructions 230 × 140 × 3 mm · straight PETG; 4 x M3 x 10 into the pod's inserts 2 mm walls, 40 % gyroid	☐
☐	print/rig/coupon_c3_wheel	25	coupon_c3_wheel	pa6-cf	2	39	2.6	Placement and instructions 30 × 30 × 21 mm · straight PA6-CF, 100 %, dried, flat (C3: M6 x 30 on a DIN 433 at 1.8 N m into a 5 mm steel plate) 100 %	☐
☐	print/rig/press_base	26	press_base	petg	1	48	1.9	Placement and instructions 73 × 20 × 50 mm · straight PETG, 6 walls, 40 % gyroid (B3) 2 mm walls, 40 % gyroid	☐
☐	print/rig/coupon_frame_pins_m01	27	coupon_frame_pins_m01	pa6-cf	1	29	1.9	Placement and instructions 46 × 28 × 24 mm · straight PA6-CF, 100 %, dried; the seat at -0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_frame_pins_nom	28	coupon_frame_pins_nom	pa6-cf	1	28	1.9	Placement and instructions 46 × 28 × 24 mm · straight PA6-CF, 100 %, dried; the seat at 0 mm on the leg's 100 %	☐
☐	print/rig/coupon_frame_pins_p01	29	coupon_frame_pins_p01	pa6-cf	1	28	1.9	Placement and instructions 46 × 28 × 24 mm · straight PA6-CF, 100 %, dried; the seat at +0.1 mm on the leg's 100 %	☐
☐	print/rig/tof_arm	30	tof_arm	petg	1	46	1.8	Placement and instructions 113 × 85 × 50 mm · straight PETG, white (the ToF's target); 2 x M4 x 10 to the plate 2 mm walls, 40 % gyroid	☐

Done	Part ID	Order	Part	Material	Quantity	Grams	Hours	Print setup	Printed
☐	print/rig/coupon_hub_stub_p01	31	coupon_hub_stub_p01	pa6-cf	1	27	1.8	Placement and instructions 64 × 64 × 23 mm · straight PA6-CF, 100 %, dried; the seat at +0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_knee_eye_m01	32	coupon_knee_eye_m01	pa6-cf	1	25	1.7	Placement and instructions 40 × 40 × 22 mm · straight PA6-CF, 100 %, dried; the seat at -0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_hub_stub_m01	33	coupon_hub_stub_m01	pa6-cf	1	26	1.7	Placement and instructions 64 × 64 × 23 mm · straight PA6-CF, 100 %, dried; the seat at -0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_hub_stub_nom	34	coupon_hub_stub_nom	pa6-cf	1	27	1.7	Placement and instructions 64 × 64 × 23 mm · straight PA6-CF, 100 %, dried; the seat at 0 mm on the leg's 100 %	☐
☐	print/rig/coupon_frame_ring_m01	35	coupon_frame_ring_m01	pa6-cf	1	25	1.6	Placement and instructions 90 × 90 × 14 mm · straight PA6-CF, 100 %, dried; the seat at -0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_frame_ring_p01	36	coupon_frame_ring_p01	pa6-cf	1	25	1.6	Placement and instructions 90 × 90 × 14 mm · straight PA6-CF, 100 %, dried; the seat at +0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_knee_eye_nom	37	coupon_knee_eye_nom	pa6-cf	1	25	1.6	Placement and instructions 40 × 40 × 22 mm · straight PA6-CF, 100 %, dried; the seat at 0 mm on the leg's 100 %	☐
☐	print/rig/coupon_knee_eye_p01	38	coupon_knee_eye_p01	pa6-cf	1	25	1.6	Placement and instructions 40 × 40 × 22 mm · straight PA6-CF, 100 %, dried; the seat at +0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_apex_eye_m01	39	coupon_apex_eye_m01	pa6-cf	1	25	1.6	Placement and instructions 90 × 90 × 13 mm · straight PA6-CF, 100 %, dried; the seat at -0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_apex_eye_nom	40	coupon_apex_eye_nom	pa6-cf	1	25	1.6	Placement and instructions 90 × 90 × 13 mm · straight PA6-CF, 100 %, dried; the seat at 0 mm on the leg's 100 %	☐
☐	print/rig/coupon_apex_eye_p01	41	coupon_apex_eye_p01	pa6-cf	1	25	1.6	Placement and instructions 90 × 90 × 13 mm · straight PA6-CF, 100 %, dried; the seat at +0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_hub_face_m01	42	coupon_hub_face_m01	pa6-cf	1	21	1.4	Placement and instructions 72 × 72 × 6 mm · straight PA6-CF, 100 %, dried; the seat at -0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_hub_face_nom	43	coupon_hub_face_nom	pa6-cf	1	21	1.3	Placement and instructions 72 × 72 × 6 mm · straight PA6-CF, 100 %, dried; the seat at 0 mm on the leg's 100 %	☐
☐	print/rig/coupon_hub_face_p01	44	coupon_hub_face_p01	pa6-cf	1	21	1.3	Placement and instructions 72 × 72 × 6 mm · straight PA6-CF, 100 %, dried; the seat at +0.1 mm on the leg's 100 %	☐

Done	Part ID	Order	Part	Material	Quantity	Grams	Hours	Print setup	Printed
☐	print/rig/press_slider	45	press_slider	petg	1	29	1.1	Placement and instructions 73 × 10 × 70 mm · straight PETG, 6 walls, 40 % gyroid (B3) 2 mm walls, 40 % gyroid	☐
☐	print/rig/coupon_knee_boss_m01	46	coupon_knee_boss_m01	pa6-cf	1	17	1.1	Placement and instructions 63 × 51 × 14 mm · straight PA6-CF, 100 %, dried; the seat at -0.1 mm on the leg's 100 %	☐
☐	print/rig/coupon_knee_boss_nom	47	coupon_knee_boss_nom	pa6-cf	1	17	1.1	Placement and instructions 63 × 51 × 14 mm · straight PA6-CF, 100 %, dried; the seat at 0 mm on the leg's 100 %	☐
☐	print/rig/coupon_knee_boss_p01	48	coupon_knee_boss_p01	pa6-cf	1	16	1.1	Placement and instructions 63 × 51 × 14 mm · straight PA6-CF, 100 %, dried; the seat at +0.1 mm on the leg's 100 %	☐
☐	print/rig/sleeve_anvil	49	sleeve_anvil	pa6-cf	1	13	0.9	Placement and instructions 30 × 18 × 30 mm · straight PA6-CF, 100 %, face down (B3) 100 %	☐
☐	print/rig/pendant_clip	50	pendant_clip	petg	1	18	0.7	Placement and instructions 76 × 44 × 22 mm · straight PETG; 2 x M4 x 10 into the pod's inserts 2 mm walls, 40 % gyroid	☐
☐	print/rig/ballast_spacer	51	ballast_spacer	petg	2	17	0.7	Placement and instructions 40 × 9 × 40 mm · straight PETG, 100 %; on each ballast post against the plate 100 %	☐
☐	print/rig/load_cell_mount	52	load_cell_mount	petg	4	17	0.7	Placement and instructions 42 × 42 × 6 mm · straight PETG; under the load plate's corners (stand and squat only) 2 mm walls, 40 % gyroid	☐
☐	print/rig/bearing_disc	53	bearing_disc	pa6-cf	3	6	0.4	Placement and instructions 20 × 20 × 6 mm · straight PA6-CF, 100 %, flat (B4: the link under an M5 head) 100 %	☐
☐	print/rig/tof_mount	54	tof_mount	petg	1	6	0.2	Placement and instructions 52 × 32 × 9 mm · straight PETG; the VL53L1X on 2 x M2.5 x 8, two wood screws into the deck 2 mm walls, 40 % gyroid	☐